Introduction to Nerual Networks

Exercise 1

Consider the simple feedforward neural network in figure 1. The hidden layer's activation function is the rectified linear unit function (ReLU), and the activation function for the output layer is the sigmoid function. For a data point y_n, assume that we use the cross-entropy loss for the output $o \in [0, 1]$ (o represents the probability $\mathrm{P}(1 \mid x, w)$):

$$L_n(w) = -y_n \ln(o) - (1 - y_n) \ln(1 - o)$$

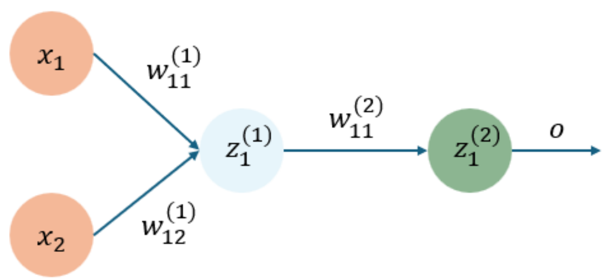

Figure 1: Simple feedforward neural network

(a) Write the gradients of the loss $L_n(w)$ with respect to the weights in the networks using back-propagation.

(b) Choose one training example (pick an $x \in \mathbb{R}^2$ and a $y \in \{1, 0\}$) and perform one update step for the weights with a suitable step size.

Exercise 2

Choose a dataset to which you want to apply a neural network. You can choose any dataset of interest; for example, you might consider:

- The CO2 data.
- The breast cancer data.
- The MNIST handwritten digit dataset.
- The CIFAR-10 image classification dataset.
- A dataset from the UCI Machine Learning Repository (e.g., the Iris dataset).

You are encouraged to implement your neural network using one of the following frameworks:

- MATLAB Neural Network Toolbox (`https://se.mathworks.com/help/deeplearning/build-deep-neural-networks.html`)
- PyTorch (`https://pytorch.org/tutorials/beginner/basics/quickstart_tutorial.html`)
- TensorFlow (`https://www.tensorflow.org/tutorials/quickstart/beginner`)
- Flux (Julia) (`https://fluxml.ai/Flux.jl/stable/guide/models/quickstart/`)

Your task is to:

(a) Load and preprocess (if needed) the chosen dataset.

(b) Design an appropriate neural network architecture for your dataset. Specify:

   - The number of layers and neurons per layer
   - The activation functions used (e.g., ReLU, sigmoid).
   - Any regularization techniques (dropout, weight decay, etc.)

   Justify your choices as much as you can based on the dataset's characteristics and the dataset.

(c) Train your neural network on the preprocessed training data.

   - Specify the loss function used (e.g., cross-entropy for classification)
   - Choose an optimizer (e.g., SGD, Adam) and note any important hyperparameters (learning rate).

(d) Evaluate the performance of your trained network on the test set.